LAB EXPERIMENT RECORD

Set 25: Website Traffic Analysis

Department	CSE-DS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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Day	Tuesday

Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

1. Line Chart in Tableau

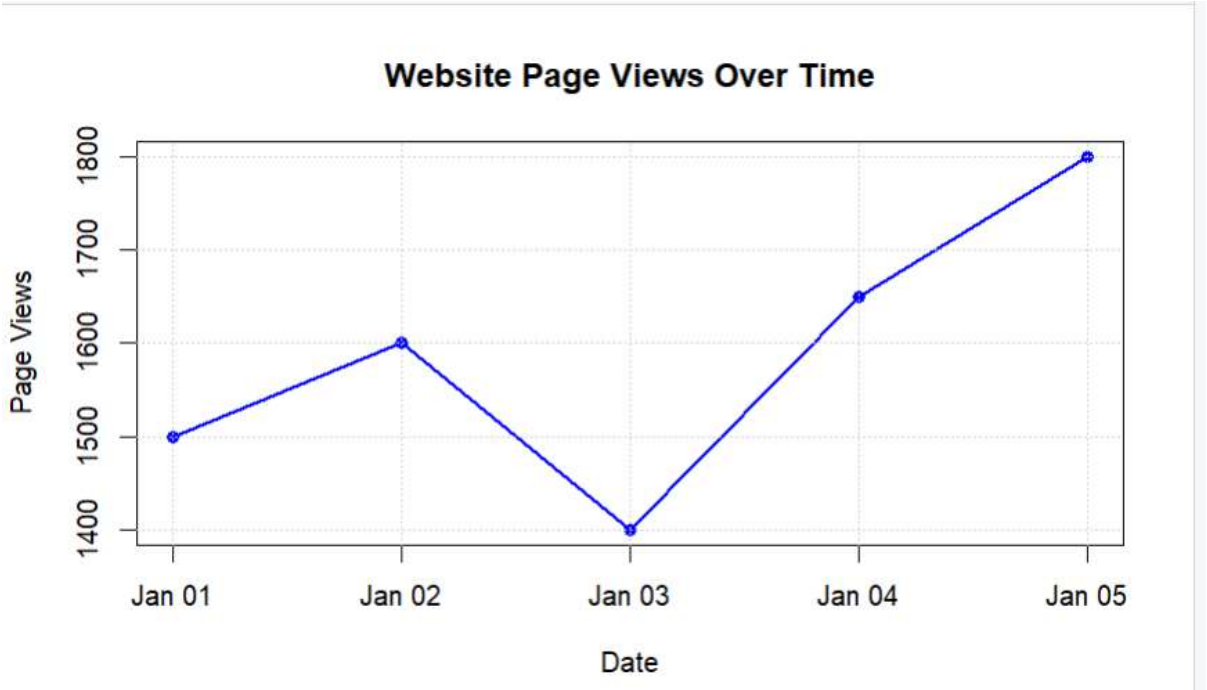
Aim

To create a line chart in Tableau to visualize the trend in page views over time and label the axes and title the chart.

Tableau Procedure

1. Open Tableau and import the Website Traffic dataset.
2. Drag Date to the Columns shelf.
3. Drag Page Views to the Rows shelf.
4. Select Line Chart from the Marks card.
5. Add the title "Website Page Views Over Time".
6. Label the X-axis as "Date" and the Y-axis as "Page Views".
7. Display the data labels if required.
8. Save the worksheet.

Output



A line chart showing the trend in website page views from January 1 to January 5, 2023.

Observation

Page views increased overall from 1500 to 1800. The lowest traffic was recorded on January 3, while the highest traffic was recorded on January 5.

2. Bar Chart in R – Top N Days with Highest Click-through Rates

Aim

To generate a bar chart in R showing the top N days with the highest click-through rates and label the chart elements.

R Code

R

```
date <- c("Jan 1", "Jan 2", "Jan 3", "Jan 4", "Jan 5")

ctr <- c(2.3, 2.7, 2.0, 2.4, 2.6)

traffic <- data.frame(Date = date, CTR = ctr)

top_days <- traffic[order(-traffic$CTR), ][1:3, ]

barplot(top_days$CTR,

        names.arg = top_days$Date,

        main = "Top 3 Days by Click-through Rate",

        xlab = "Date",

        ylab = "Click-through Rate (%)",

        col = "skyblue",

        border = "black",

        ylim = c(0, 3))

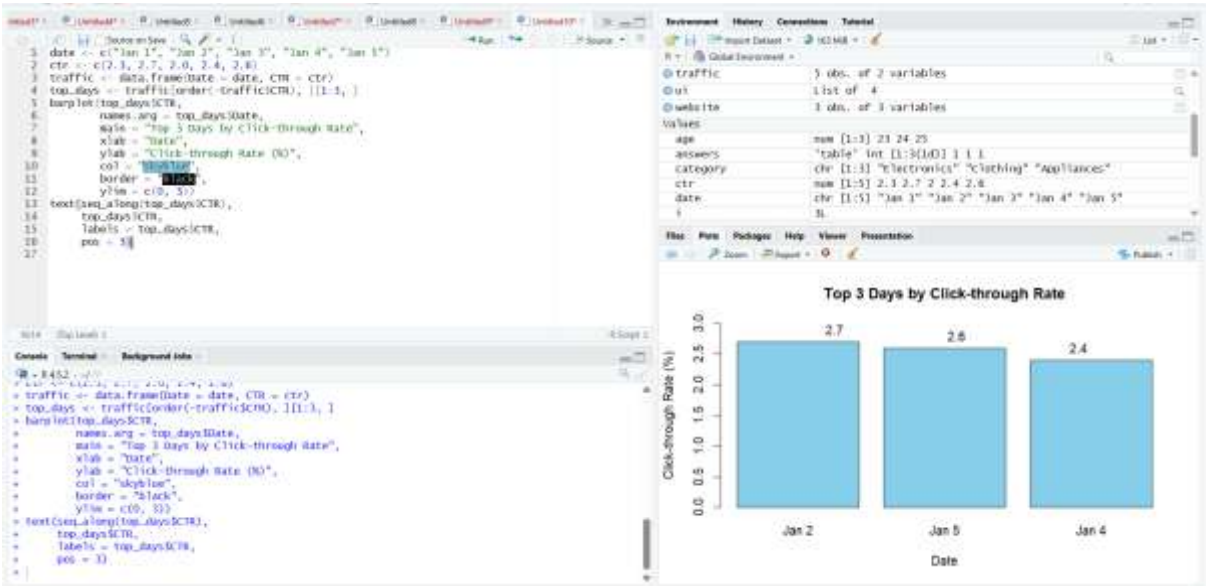
text(seq_along(top_days$CTR),

     top_days$CTR,

     labels = top_days$CTR,

     pos = 3)
```

Output



A bar chart displaying the three days with the highest click-through rates.

Observation

January 2 recorded the highest click-through rate of 2.7%, followed by January 5 at 2.6% and January 4 at 2.4%.

3. Stacked Area Chart in R – User Interactions

Aim

To build a stacked area chart in R to display the distribution of user interactions such as likes, shares, and comments on the website.

R Code

R

```
date <- c("Jan 1", "Jan 2", "Jan 3", "Jan 4", "Jan 5")

likes <- c(120, 150, 100, 140, 180)

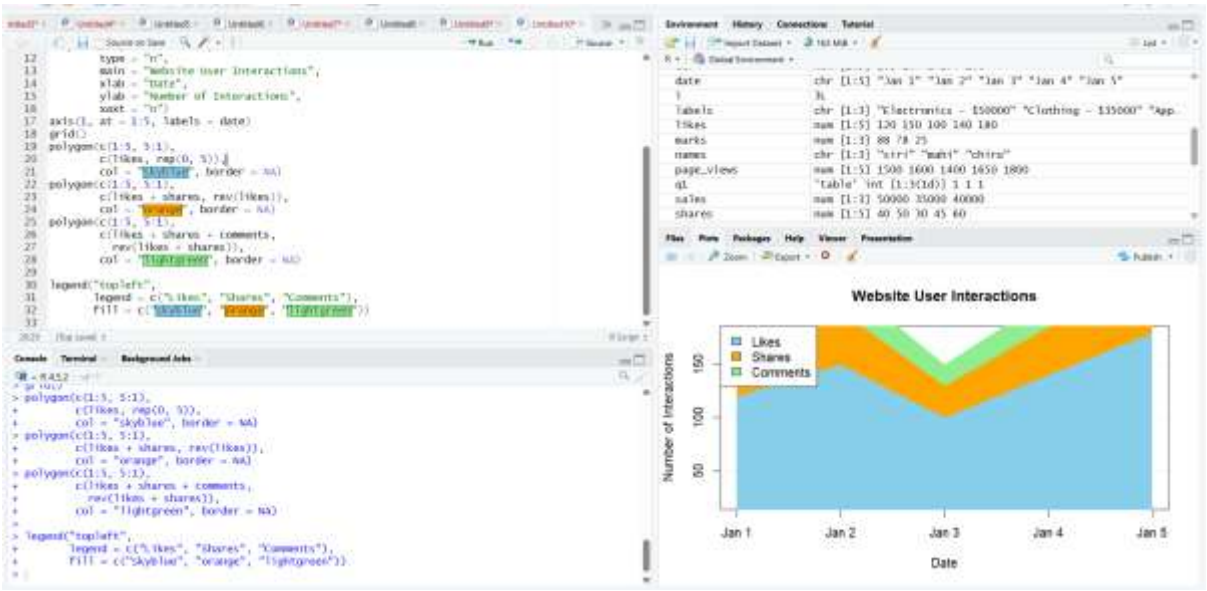
shares <- c(40, 50, 30, 45, 60)

comments <- c(25, 30, 20, 28, 35)
```

```
interactions <- data.frame(
  Likes = likes,
  Shares = shares,
  Comments = comments
)
matplot(1:5, interactions,
        type = "n",
        main = "Website User Interactions",
        xlab = "Date",
        ylab = "Number of Interactions",
        xaxt = "n")
axis(1, at = 1:5, labels = date)
grid()
polygon(c(1:5, 5:1),
        c(likes, rep(0, 5)),
        col = "skyblue", border = NA)
polygon(c(1:5, 5:1),
        c(likes + shares, rev(likes)),
        col = "orange", border = NA)
polygon(c(1:5, 5:1),
        c(likes + shares + comments,
          rev(likes + shares)),
        col = "lightgreen", border = NA)
```

```
legend("topleft",
       legend = c("Likes", "Shares", "Comments"),
       fill = c("skyblue", "orange", "lightgreen"))
```

Output



A stacked area chart showing the distribution of likes, shares, and comments over five days.

Observation

Likes represent the largest number of interactions, followed by shares and comments. The interaction values are assumed for demonstration because the original dataset does not contain user interaction data.

4. Tableau Dashboard – Traffic Sources and Page Views

Aim

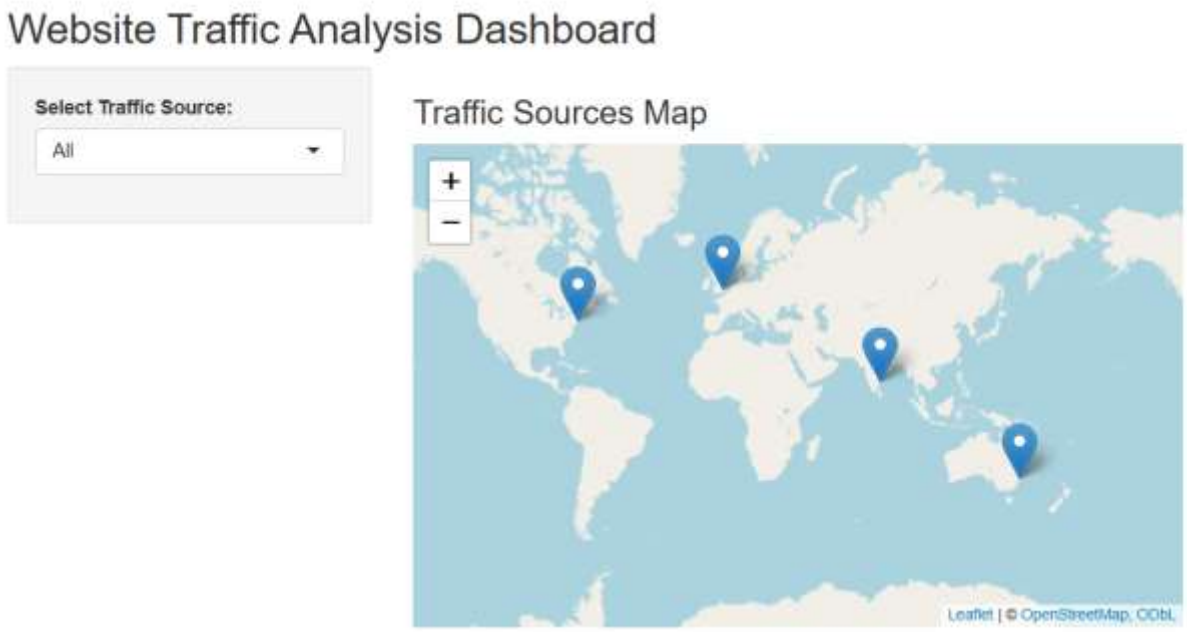
To create a Tableau dashboard with an interactive map showing traffic sources and a bar chart displaying page views by source.

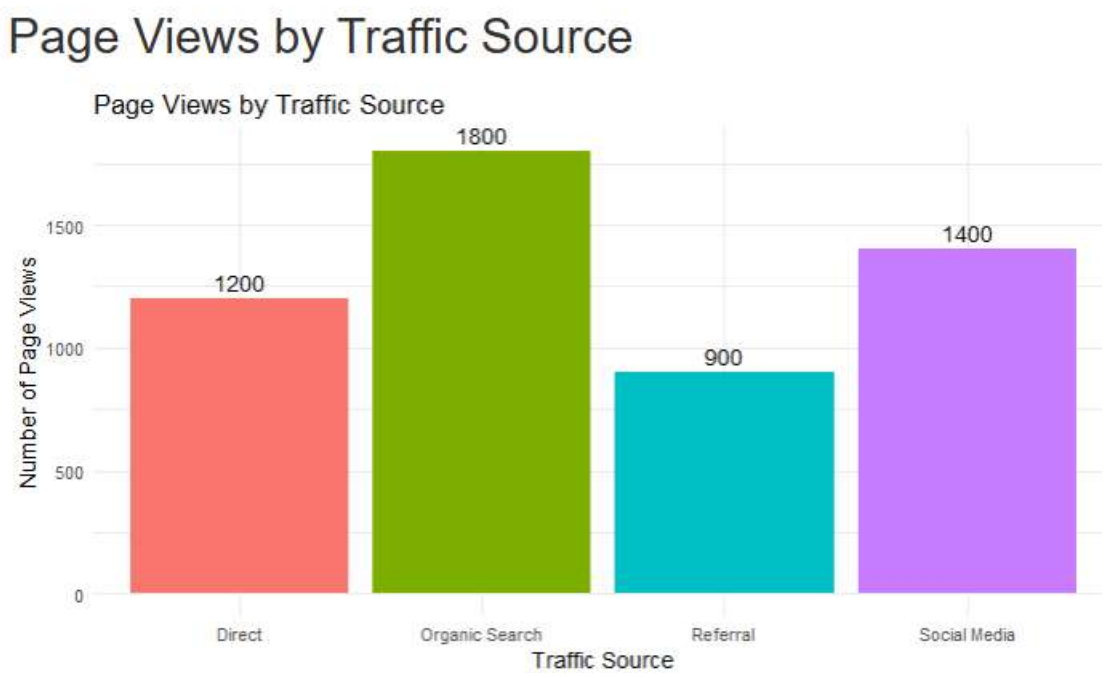
Tableau Procedure

1. Open Tableau and import the Website Traffic dataset.

2. Create a separate traffic source dataset containing Source, Country, Latitude, Longitude, and Page Views.
3. Use sample sources such as Organic Search, Social Media, Direct, and Referral.
4. Drag Longitude to Columns and Latitude to Rows to create a map.
5. Add Traffic Source to Color and Page Views to Size in the Marks card.
6. Create a bar chart using Traffic Source and Page Views.
7. Select Dashboard → New Dashboard.
8. Drag the map and bar chart into the dashboard.
9. Add suitable titles, labels, and colors.
10. Enable dashboard filter actions so selecting a traffic source filters the bar chart.
11. Save the dashboard.

Output





An interactive dashboard displaying website traffic sources on a map and comparing page views by traffic source.

Observation

The dashboard helps identify the geographical distribution of traffic sources and compare the number of page views contributed by each source. The traffic source dataset is assumed for demonstration because the original dataset does not contain source or location details.

Result

The Website Traffic dataset was analyzed using a Tableau line chart, an R bar chart, an R stacked area chart, and a Tableau interactive dashboard. The visualizations help identify page-view trends, compare click-through rates, understand user interactions, and explore traffic sources.